

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.0621151242148579, median 0.9215329801936646, std: 0.7018608575987746
Reprojection error (cam1): mean 1.0145274758672858, median 0.8545509129283693, std: 0.7036384703760202
Gyroscope error (imu0): mean 1.7169387643264098, median 1.4894117988490234, std: 0.9847955168917015
Accelerometer error (imu0): mean 0.44957321394670774, median 0.39361384780831266, std: 0.274934230530444

Residuals

Reprojection error (cam0) [px]: mean 1.0621151242148579, median 0.9215329801936646, std: 0.7018608575987746
Reprojection error (cam1) [px]: mean 1.0145274758672858, median 0.8545509129283693, std: 0.7036384703760202
Gyroscope error (imu0) [rad/s]: mean 0.0171693876432641, median 0.014894117988490232, std: 0.009847955168917014
Accelerometer error (imu0) [m/s²]: mean 0.17982928557868308, median 0.15744553912332504, std: 0.10997369221217759

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.00218501  0.02784515  0.99960986  0.01191828]
 [-0.00266759  0.99960853 -0.02785095  0.16813082]
 [-0.99999405 -0.00272741 -0.00210988 -0.1149622 ]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[-0.00218501 -0.00266759 -0.99999405 -0.11448697]
 [ 0.02784515  0.99960853 -0.00272741 -0.16871042]
 [ 0.99960986 -0.02785095 -0.00210988 -0.00747359]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.0002875771984465692

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.01564593 0.02338682 0.99960405 -0.03675999]
[-0.006304 0.99970886 -0.0232906 0.16793131]
[-0.99985772 -0.0059371 0.01578881 -0.11623929]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.01564593 -0.006304 -0.99985772 -0.11458896]
[0.02338682 0.99970886 -0.0059371 -0.16771284]
[0.99960405 -0.0232906 0.01578881 0.04249193]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.00029766560555830106

Baselines:

Baseline (cam0 to cam1):
[[0.99983109 -0.00450399 -0.01781867 -0.04996747]
[0.0045693 0.99998298 0.00362649 0.0001658]
[0.01780203 -0.00370729 0.99983466 -0.00088495]
[0. 0. 0. 1.]]
baseline norm: 0.049975578152190045 [m]

Gravity vector in target coords: [m/s²]
[0.27011565 -9.8028292 -0.00032241]

Calibration configuration
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cam0

Camera model: omni
Focal length: [2044.6753301041977, 2065.5927559749084]
Principal point: [616.3310327581959, 404.1255934969876]
Omni xi: 1.5936757817618359
Distortion model: radtan
Distortion coefficients: [2.290964717370836, -1.8671244246514953, -0.010010836103043635, 0.002831387673779597]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

cam1

Camera model: omni
Focal length: [2037.2418743359228, 2054.1495623160185]
Principal point: [634.1340582275024, 379.92908128448573]
Omni xi: 1.569995877298825
Distortion model: radtan

Distortion coefficients: [2.167702619581351, -0.9560356055374504, -0.01088770643867036, 0.012396986450845738]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

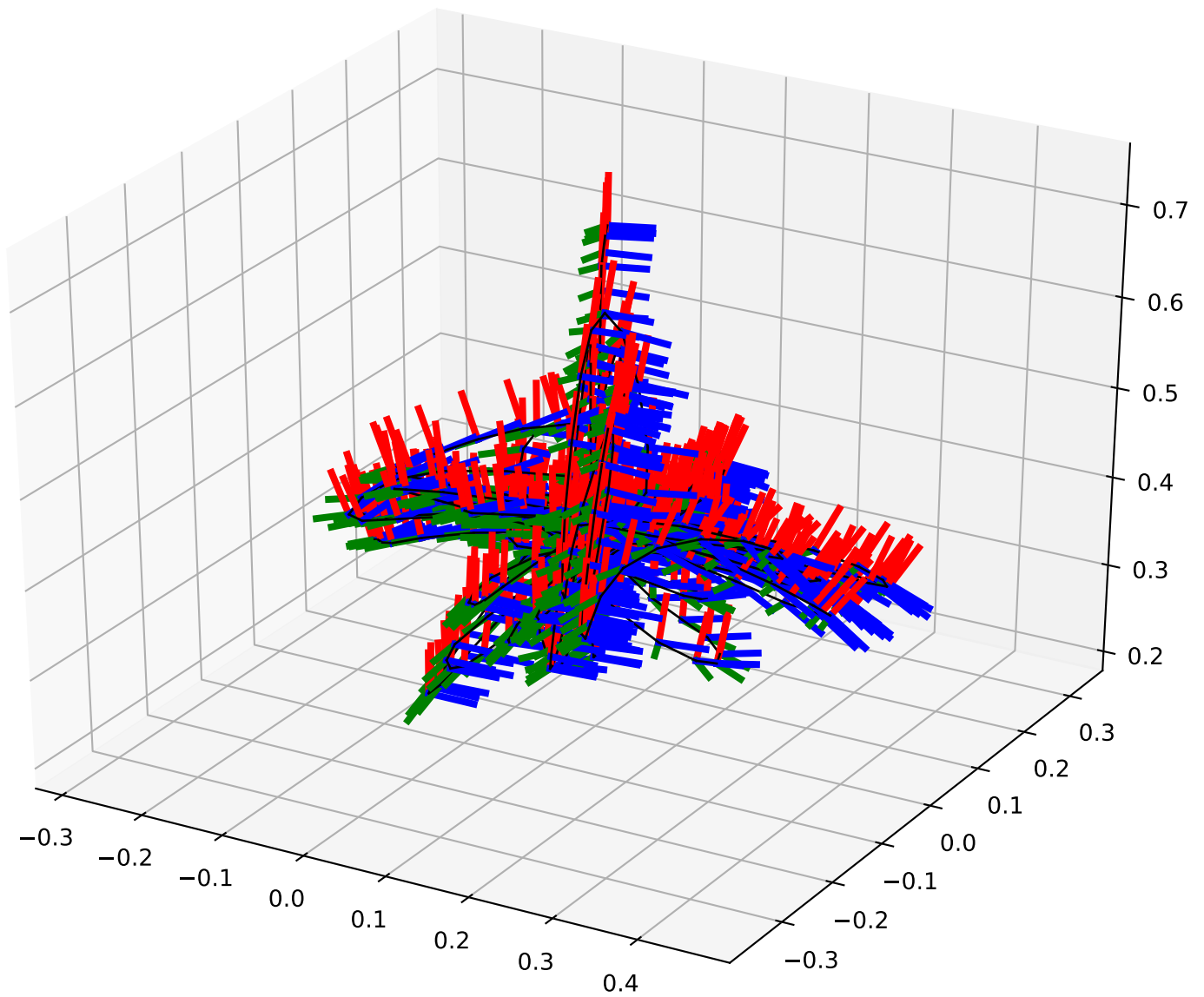
IMU configuration
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IMU0:

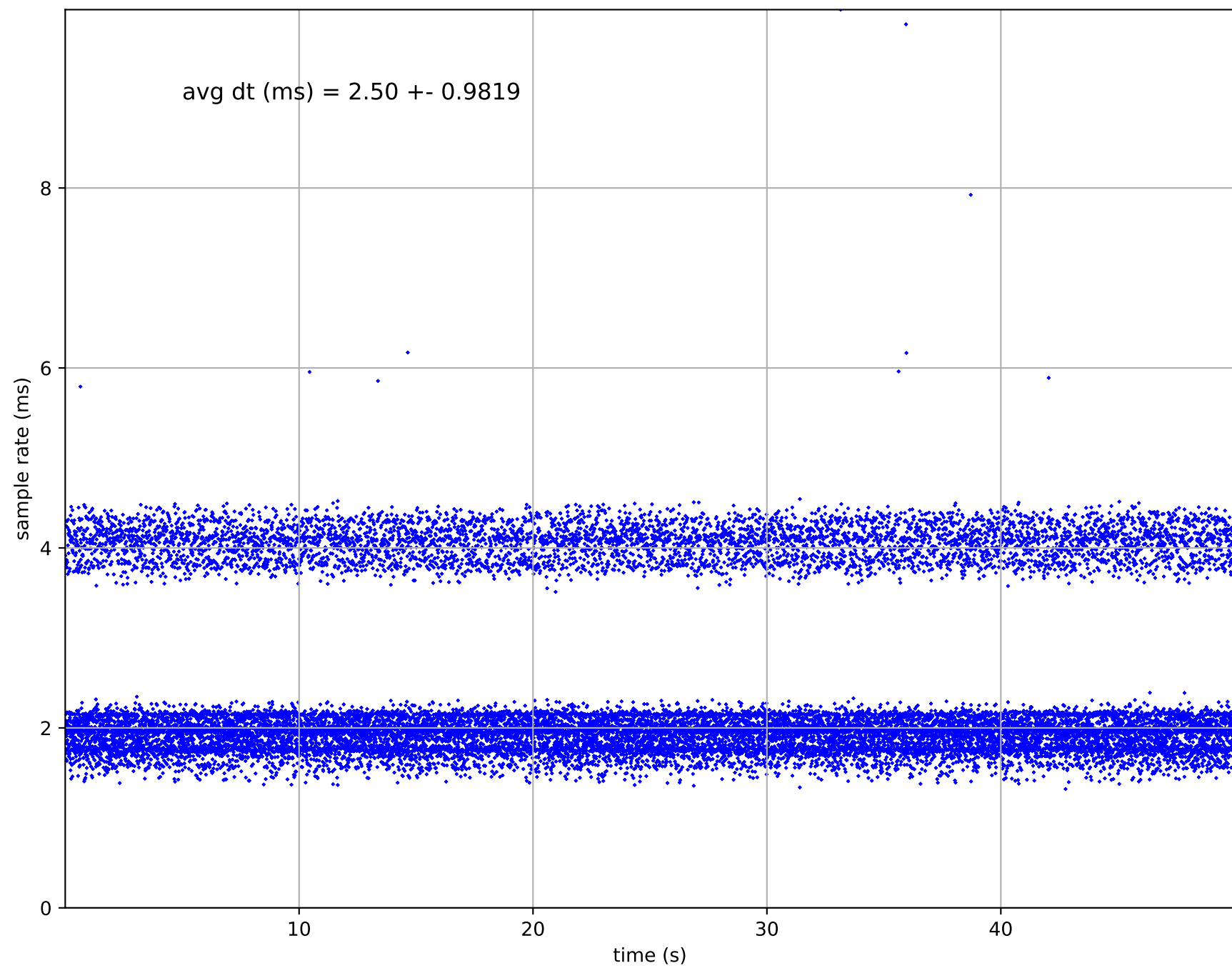
Model: calibrated
Update rate: 400
Accelerometer:
Noise density: 0.02
Noise density (discrete): 0.3999999999999997
Random walk: 0.001
Gyroscope:
Noise density: 0.0005
Noise density (discrete): 0.01
Random walk: 0.0001
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

imu0: estimated poses

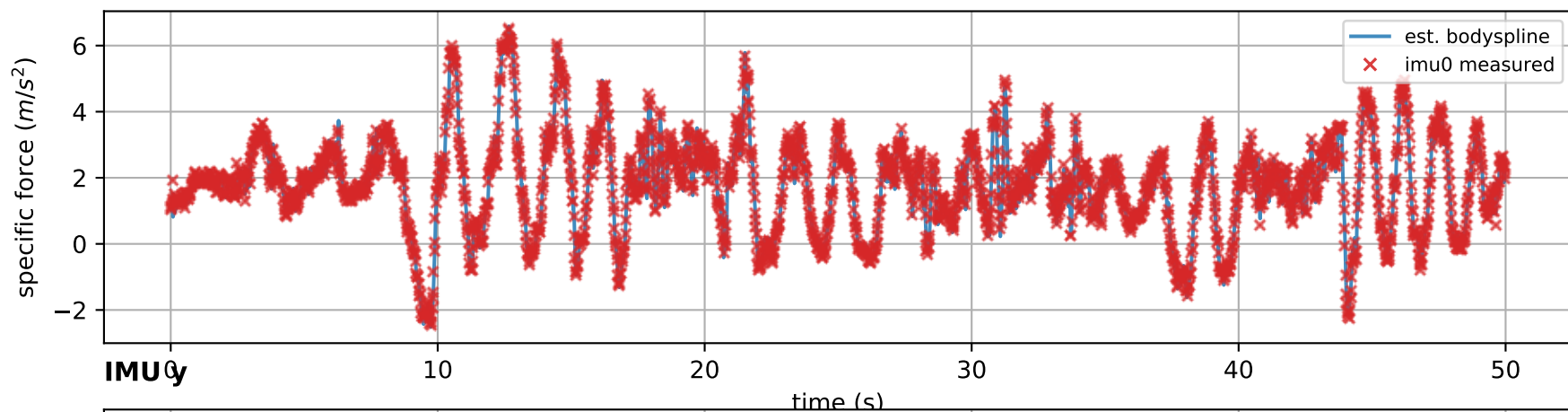


imu0: sample inertial rate

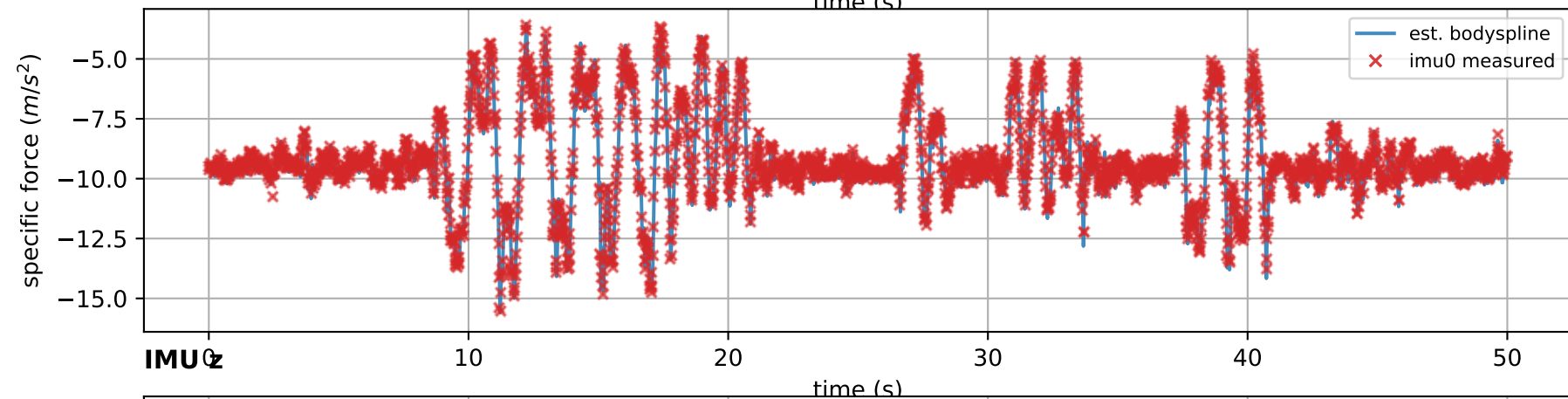


Comparison of predicted and measured specific force (imu0 frame)

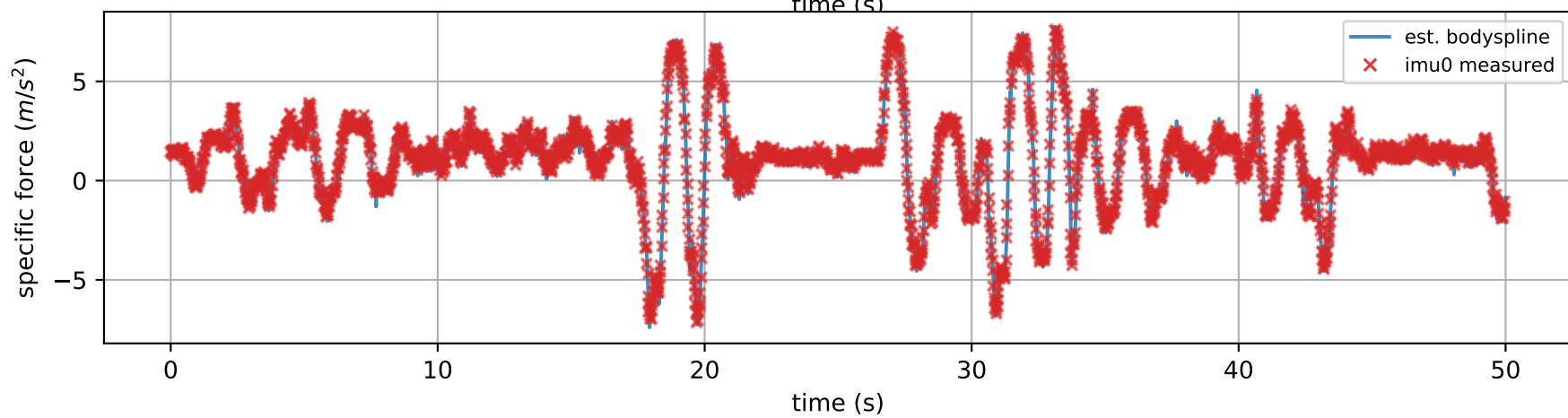
IMU x



IMU y

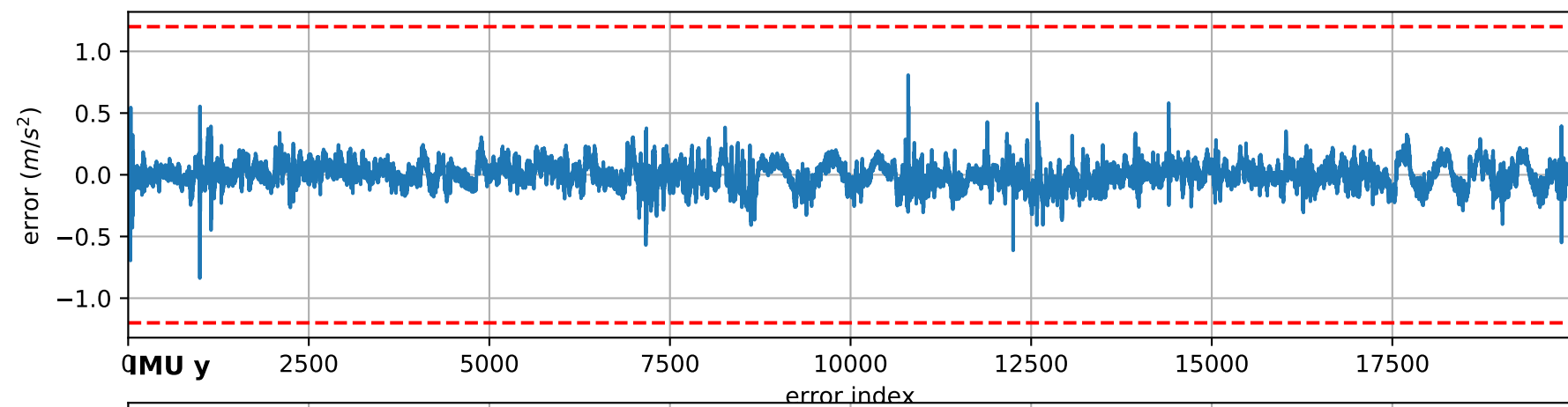


IMU z

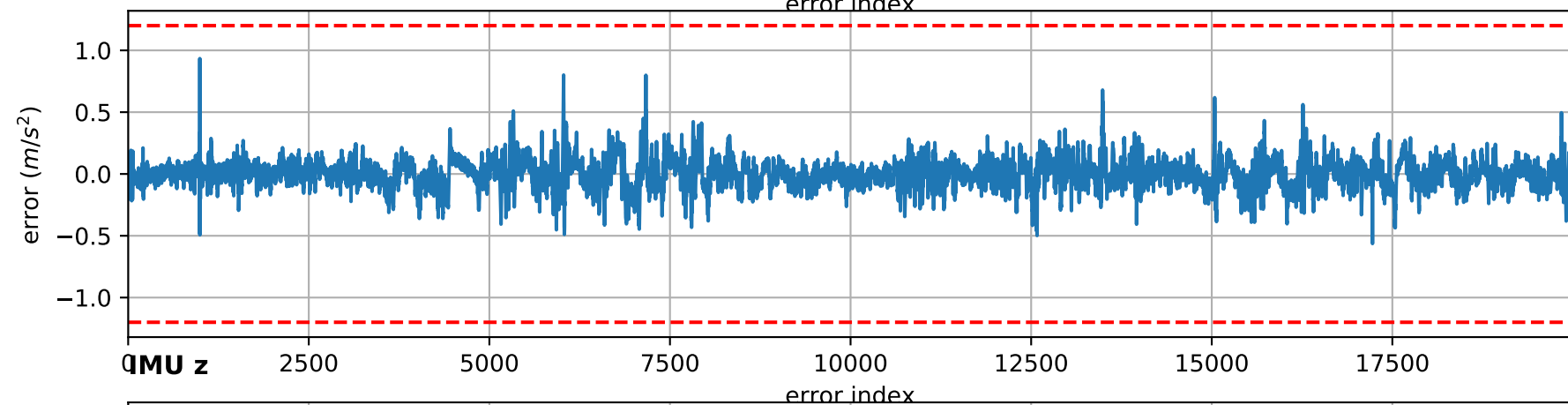


imu0: acceleration error

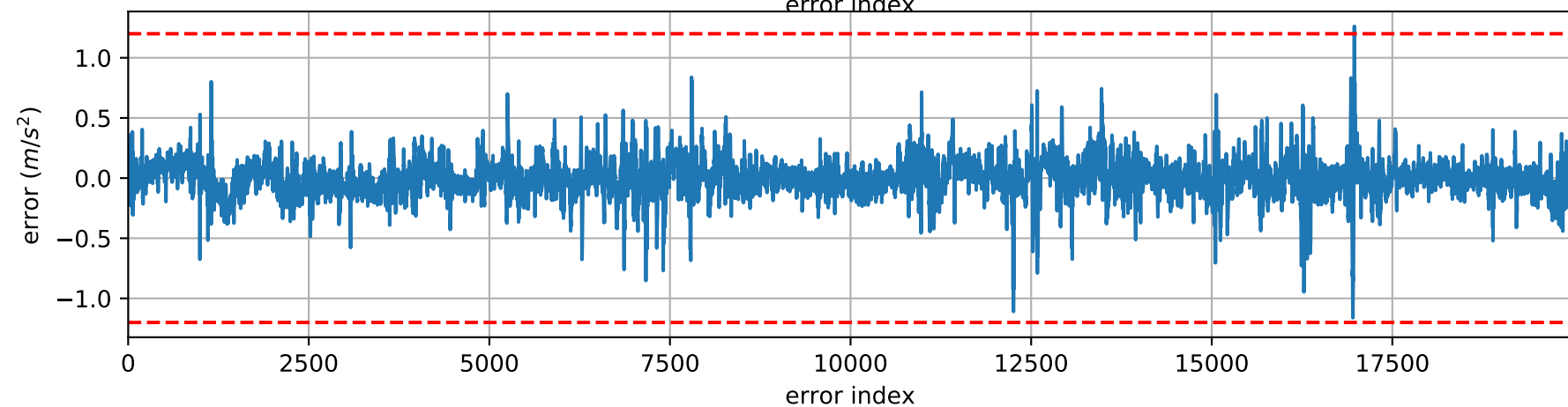
IMU x



IMU y

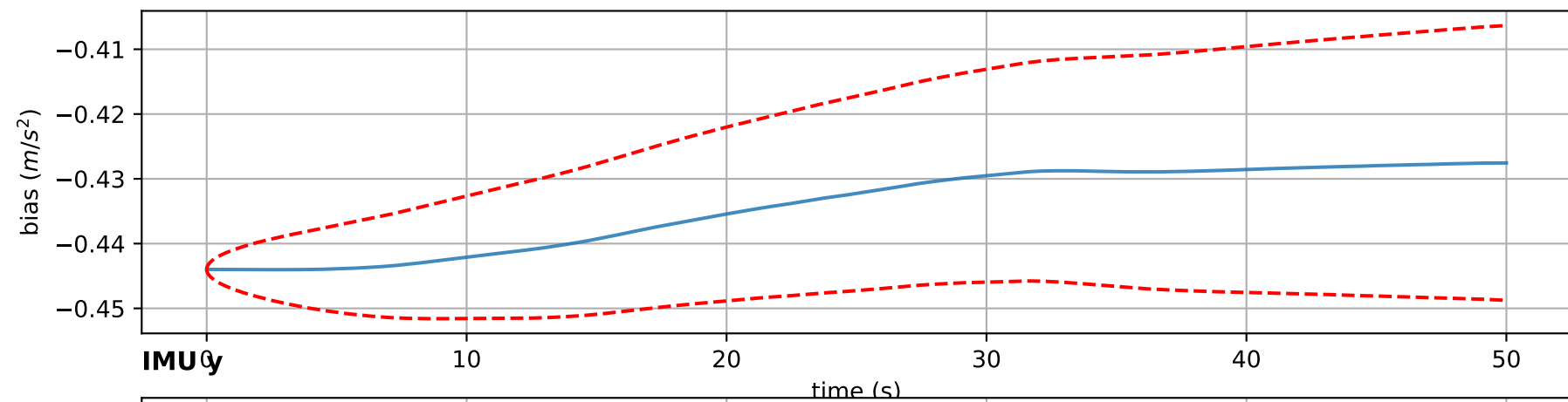


IMU z

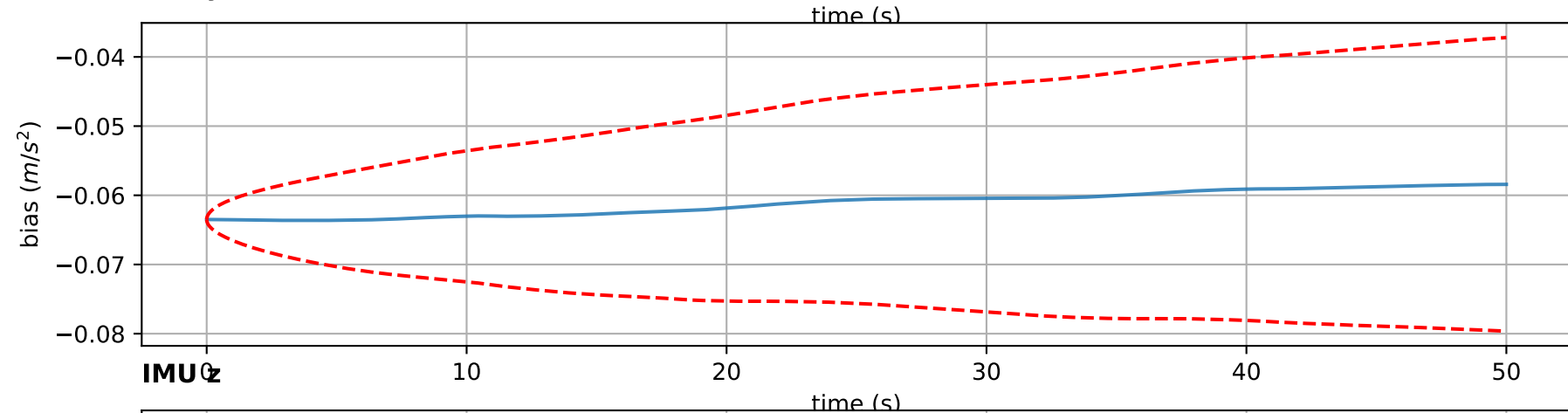


imu0: estimated accelerometer bias (imu frame)

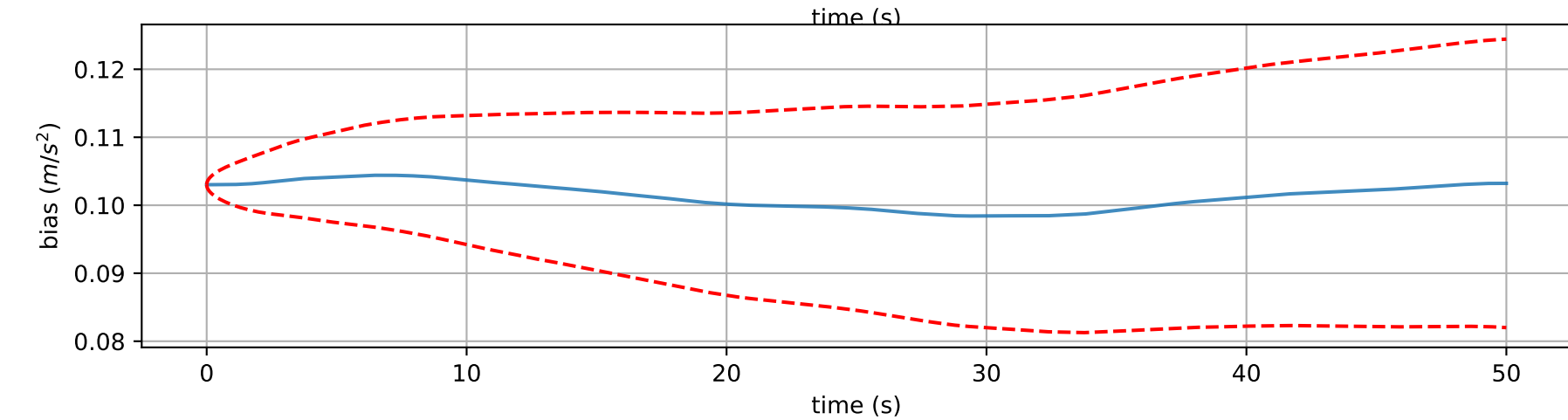
IMU x



IMU y

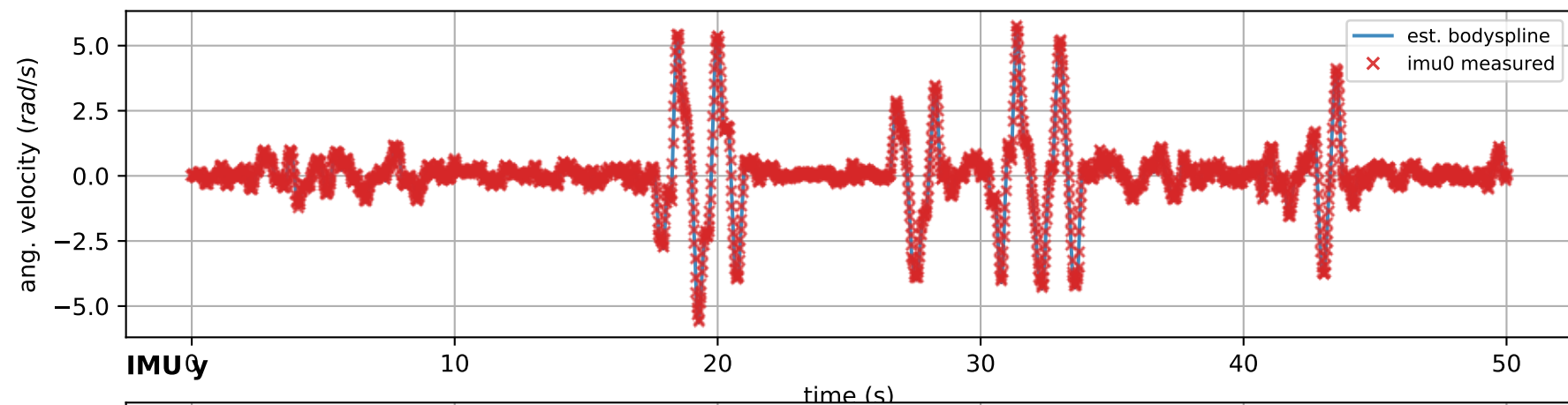


IMU z

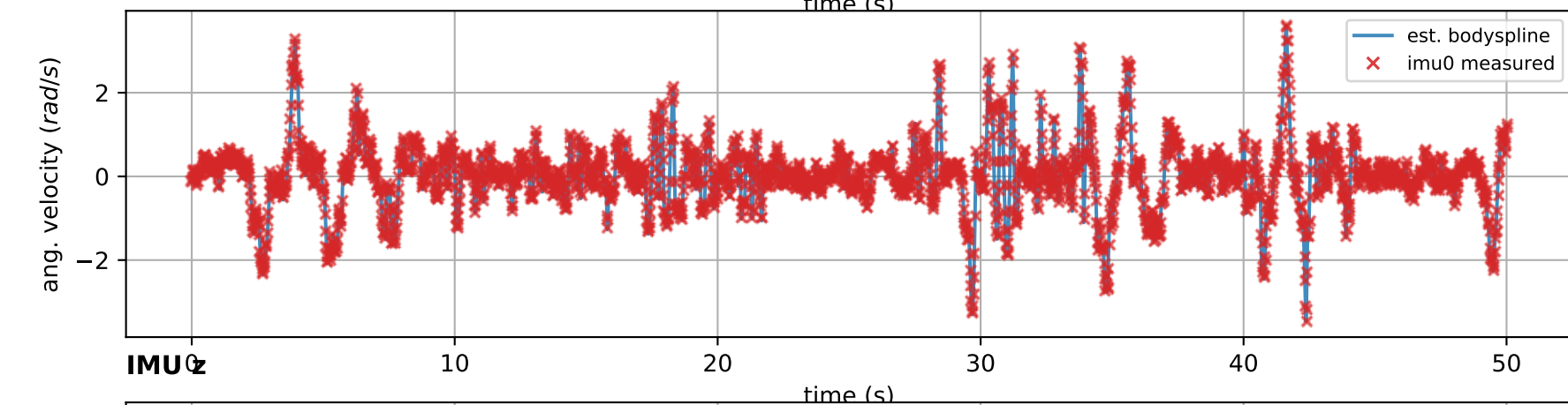


Comparison of predicted and measured angular velocities (body frame)

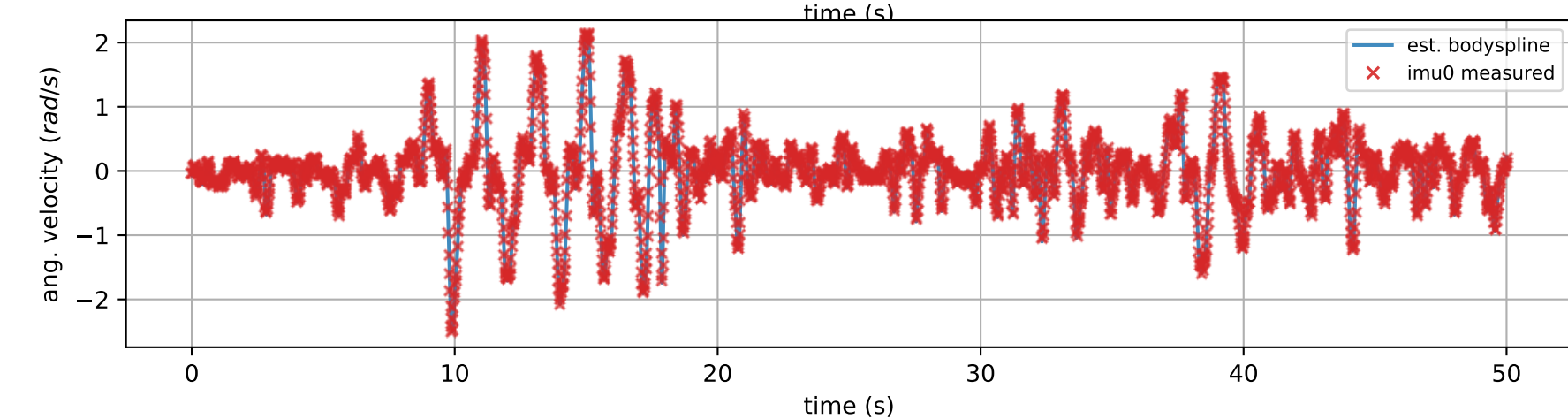
IMU x



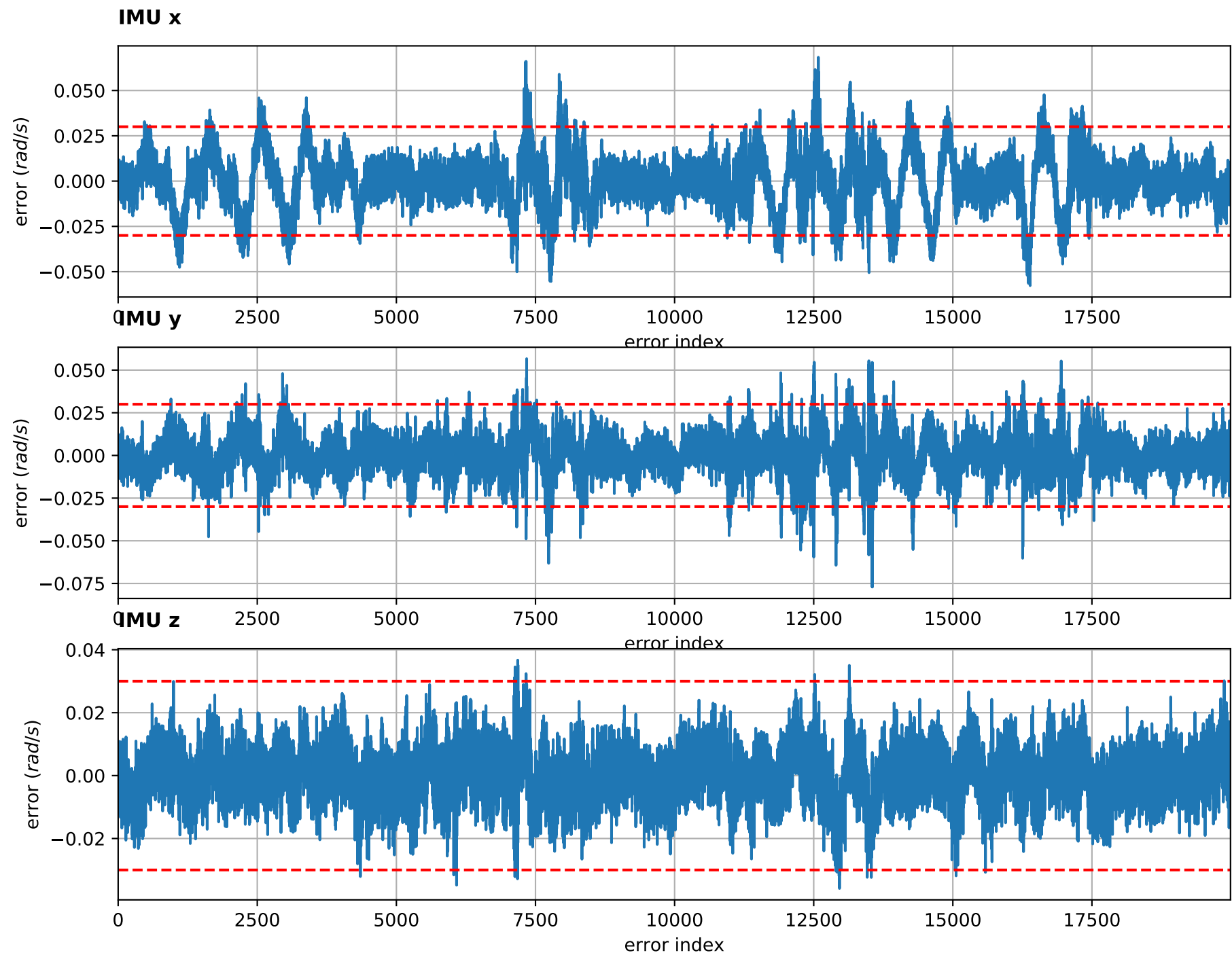
IMU y



IMU z

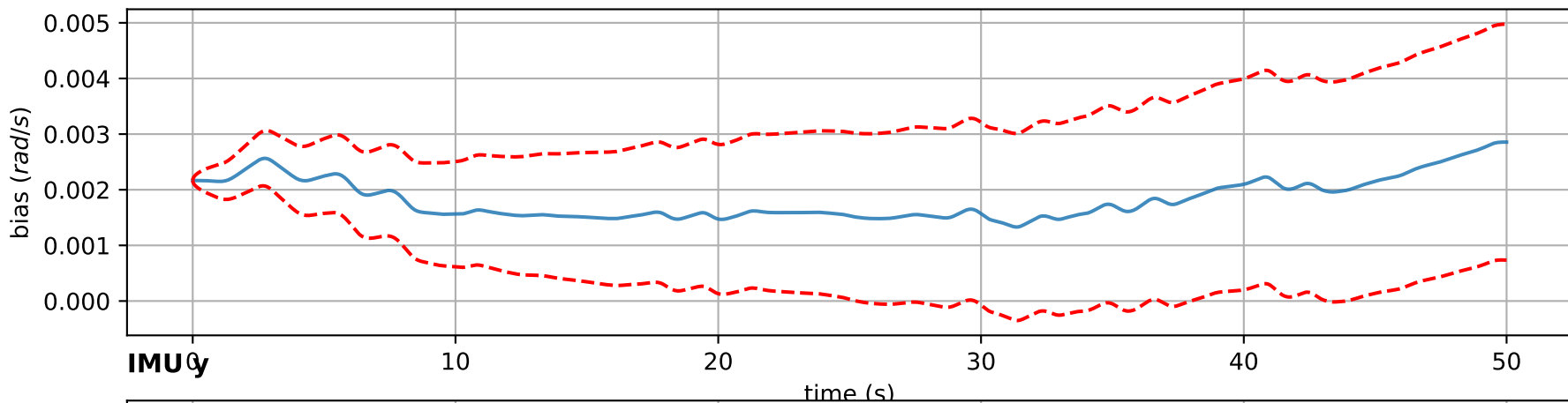


imu0: angular velocities error

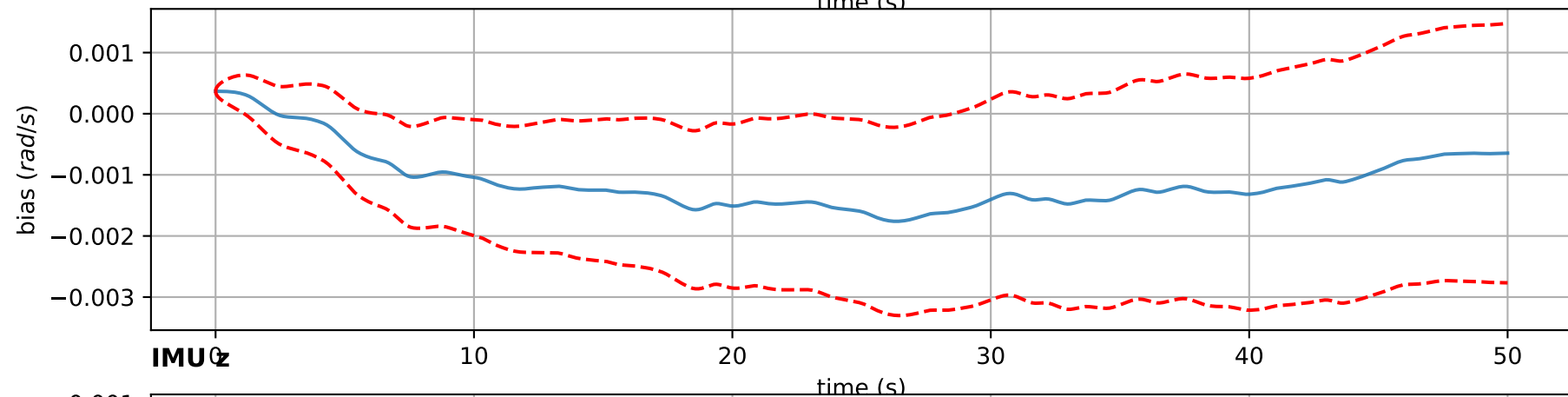


imu0: estimated gyro bias (imu frame)

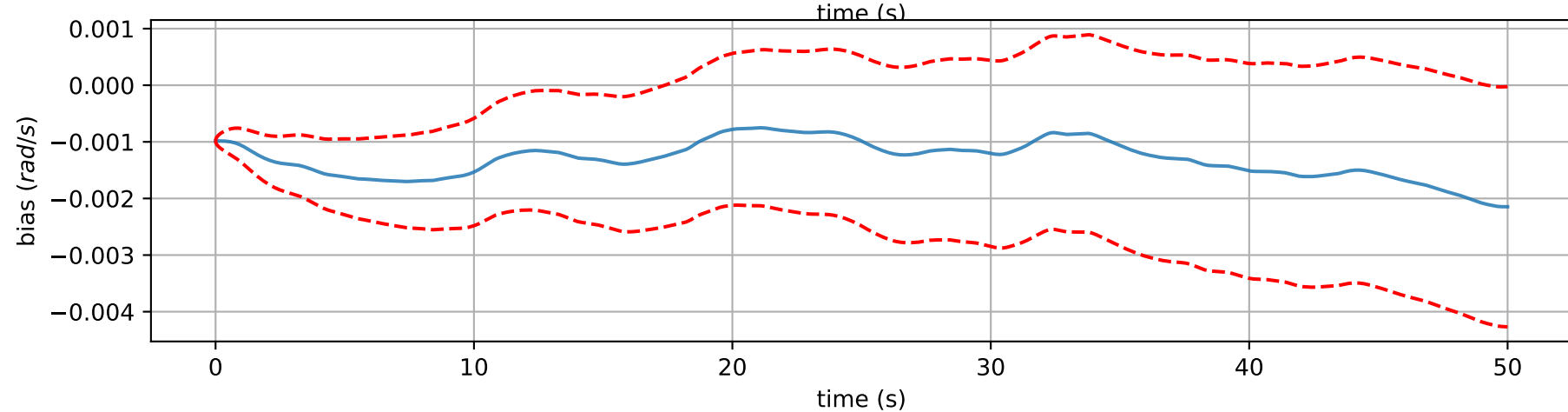
IMU x



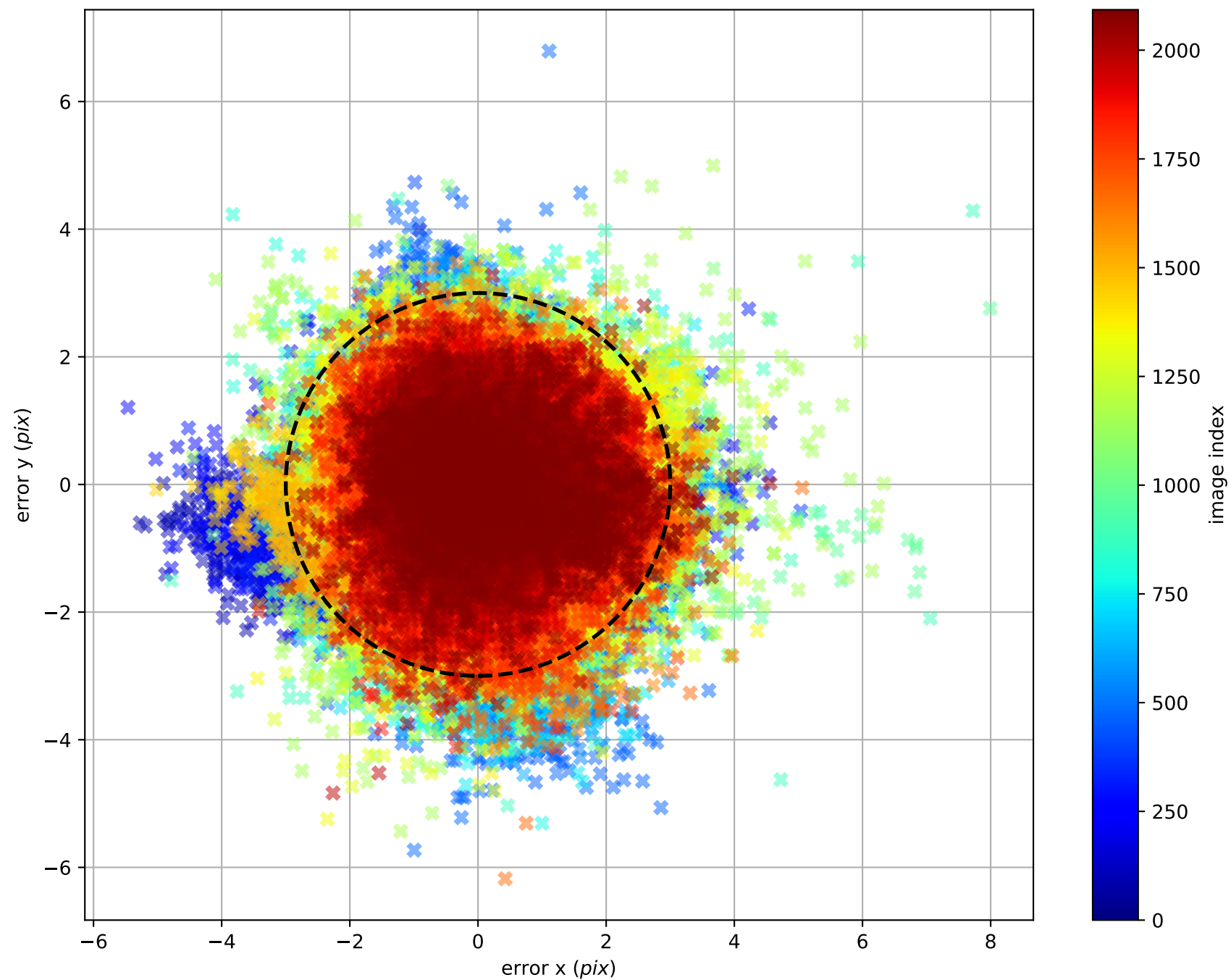
IMU y



IMU z



cam0: reprojection errors



cam1: reprojection errors

